YEAR 7 · UNIT 4 · HOMEWORK 9

Lighter on the Moon

Mathematics 4.1–4.5 and the Unit 4 review — decimals
Section F — Science 3.1 in decimals: weight, gravity and falling

Name: _____ **Class:** _____

Date given: _____ **Date due:** Friday 23 October 2026

How to do this homework

Six sections. A to E are the decimal lessons. F is the science from 3.1, and **every answer in F is a decimal**.

- **Write the calculation before you write the answer.**
- **Estimate first.** The estimate tells you where the point goes.
- **No calculator.** In F, answer in **full English sentences** where it says so.

Section A · Mathematics 4.1 — Comparing and Ordering

A1 Put each number on the line

The line goes from **5** to **6**. Every small line is **0.1**. **(a)** Draw an arrow to each number and write its letter.



A 5.2	**B** 5.9	**C** 5.45	**D** 5.05

(b) Now write the four numbers **in order of size, starting with the smallest**.

_____

A2 Which is bigger?

Compare decimals

First look at the **whole-number parts**. If they are the same, **fill with zeros** so both numbers have the same number of digits after the point, then compare. 3.3 = 3.30, and 3.30 > 3.25.

Write <, > or = in each box.

(a) 7.4 ☐ 7.38 (c) 3.05 ☐ 3.5 (e) 0.70 ☐ 0.7

(b) 0.6 ☐ 0.59 (d) 12.3 ☐ 9.99 (f) 4.125 ☐ 4.13

(g) Mr Bowen says: *"6.18 is bigger than 6.2, because 18 is bigger than 2."*

(i) Write how you **say** each number in English.

6.18: _____ 6.2: _____

(ii) Is he right? Explain. Start: **"Mr Bowen is … because …"**

A3 Who won?

Word help — the words that tell you which end

longest the biggest distance

fastest the **smallest** time. Fast runners and swimmers take **less** time

fourth number 4 in the order. *Fourth fastest*: the fourth smallest time

Long jump

Jumper	Distance (m)
A	4.8
B	4.08
C	4.75
D	4.79
E	4.5

50 m swim

Lane	Time (s)
1	31.4
2	30.95
3	31.09
4	30.9
5	31.12

(a) Write the five jumps in order of size, starting with the smallest.

_____

(b) The **longest** jump wins. Which jumper won? _____

(c) Write the five swim times in order of size, starting with the smallest.

_____

(d) Which lane was the **fastest**? _____

(e) Which lane came **fourth**? _____

(f) Which lane was the **slowest**? _____

Section B · Mathematics 4.2 — Adding and Subtracting

B1 Use a written method

Line up the points

A **written method** is in columns on paper. Write the numbers with the **points in a line**. Fill the gaps with **zeros**. Work from the right, like whole numbers. The point in the answer goes **straight down**.

(a) 5.36 + 2.7

(b) 14.8 + 3.65

(c) 0.45 + 6 + 1.9

(d) 9.4 − 3.75

(e) 12 − 4.36

(f) 1 − 0.65

B2 Use a mental method

A **mental method** is in your head. Change one number to a whole number, then give back the difference. Write the easier calculation, then the answer. The first one is done for you.

Calculation	The easier way	Answer
5.8 + 2.9	5.8 + 3 − 0.1	8.7
3.9 + 2.5		
7.3 − 1.9		
6.99 + 3.45		
12.6 − 4.98		

Section C · Mathematics 4.3–4.4 — Multiplying and Dividing

C1 Point out, point back

Multiplying a decimal

Point out: 32 × 7 = 224. Point back, **same number of decimal places**: 3.2 × 7 = 22.4
Estimate to check: 3 × 7 = 21, and 22.4 is close to 21.

Fill in the table. The first one is done for you.

Calculation	Estimate	Without the point	Answer
4.6×3	$5 \times 3 = 15$	$46 \times 3 = 138$	13.8
6.4×5			
0.7×8			
2.35×4			
8.15×2			

C2 The point goes straight up

Dividing a decimal

Use short division. The point in the answer goes **straight up**.
If a digit is too small to divide, write **0** and carry it.
Check with the **inverse calculation**: $8.24 \div 4 = 2.06$ because $2.06 \times 4 = 8.24$.

(a) $7.56 \div 3$

(b) $6.24 \div 6$

(c) $0.84 \div 7$

(d) $5.4 \div 4$

(e) $16.2 \div 5$

(f) $8.08 \div 8$

(g) Check **(b)** with the inverse calculation. _____ × _____ = _____

C3 One fact, five answers

You are given one fact. **Do not multiply again.** Use the fact, and an estimate to decide where the point goes.

$$38 \times 27 = 1026$$

(a) $3.8 \times 27 =$ _____

(b) $0.38 \times 27 =$ _____

(c) $38 \times 2.7 =$ _____

(d) $102.6 \div 27 =$ _____

(e) $10.26 \div 27 =$ _____

(f) Two of your answers are the same. Which two, and why?

Section D · Mathematics 4.5 — Making Calculations Easier

D1 Decimals as fractions

Word help

equivalent	equal in value, but written differently. 0.3 and $\frac{3}{10}$ are equivalent
tenths · hundredths	the column of the **last** digit is the bottom number

Write each decimal as a fraction, and each fraction as a decimal.

(a) 0.7 = _____ (c) 0.31 = _____ (e) $\frac{8}{100}$ = _____

(b) 0.03 = _____ (d) 0.6 = _____ (f) $\frac{51}{100}$ = _____

D2 Half of, a quarter of, a tenth of

In English, "of" often means ×

× 0.5 = **half of** = ÷ 2 × 0.25 = **a quarter of** = ÷ 4 × 0.1 = **a tenth of** = ÷ 10

Fill in the table. The first one is done for you.

Calculation	In English	Answer
48 × 0.5	half of 48	24
36 × 0.25		
70 × 0.1		
	a quarter of 2.8	
	a tenth of 4.5	

D3 Multiplying by a decimal

Divide first, then multiply

× 0.2 means ÷ 10, then × 2. × 0.03 means ÷ 100, then × 3.
So 250 × 0.2 = 25 × 2 = 50. Multiplying by a decimal **less than 1** makes a number **smaller**.

(a) 40 × 0.3 = _____ × _____ = _____ (c) 600 × 0.04 = _____ × _____ = _____

(b) 150 × 0.2 = _____ × _____ = _____ (d) 90 × 0.7 = _____ × _____ = _____

(e) $1200 \times 0.05 =$ _____ × _____ = _____ (f) $80 \times 0.6 =$ _____ × _____ = _____

D4 Times 9, times 11, and partitioning

Three quick ways

× 9: do × 10, then **take away** one lot. $3.8 \times 9 = 38 - 3.8 = 34.2$
× 11: do × 10, then **add** one lot. $3.8 \times 11 = 38 + 3.8 = 41.8$
Partitioning: split the number. $4.3 \times 6 = 4 \times 6 + 0.3 \times 6 = 24 + 1.8 = 25.8$

Write the quick way, then the answer.

(a) $4.6 \times 9 =$ _____ = _____

(b) $2.3 \times 11 =$ _____ = _____

(c) $6.2 \times 7 =$ _____ = _____

(d) $3.8 \times 5 =$ _____ = _____

D5 Mr Bowen's homework

Mr Bowen gave himself 4 out of 4. Only **one** of these is right. Find it, and correct the other three.

(a) $0.4 \times 7 = 0.28$ _____

(b) $5.09 > 5.1$ _____

(c) $7.2 \div 6 = 1.2$ _____

(d) $50 \times 0.3 = 150$ _____

Section E · Unit 4 Review — Bigger Numbers, Real Problems

E1 Two-digit numbers

Estimate, or write the table

× a 2-digit number: take the point out, multiply, then use an **estimate** to put it back.
4.7×32: $47 \times 32 = 1504$, and $5 \times 30 = 150$, so **150.4**.
÷ a 2-digit number: write its **times table** first, then use short division. The point goes straight up.

(a) 5.8×15

(b) $41.6 \div 13$

E2 From English to a calculation

Word help — the words that choose the operation

change	in a shop, the money you get **back** when you pay too much. Take away
share equally	give everyone the same amount. Divide — and count the people carefully
how much heavier	the difference. Take away
subtract 2 from 9	9 − 2. The **second** number comes **first**

Write the **calculation first**, then the answer. Give units.

(a) Mr Bowen pays for a 3.65-dollar sandwich with a 5-dollar note. How much change does he get?

Calculation: _____ **Answer:** _____

(b) A mango costs 1.20 dollars. How much do 6 mangoes cost?

Calculation: _____ **Answer:** _____

(c) Mr Bowen and two friends share a 13.50-dollar pizza equally. How much does each person pay?

Calculation: _____ **Answer:** _____

(d) A bag of rice weighs 2.4 kg. A bag of sugar weighs 0.85 kg. How much heavier is the rice?

Calculation: _____ **Answer:** _____

(e) Subtract 2.6 from 9.1.

Calculation: _____ **Answer:** _____

E3 Money has two decimal places

Money

Money is written to **2 decimal places**. 4.5 dollars is written **4.50** dollars.
10 ÷ 3 = 3.333 . . ., so three friends get **3.33** dollars each, and 1 cent is left over.

Write each one as money.

(a) 4.8 dollars _____ **(b)** 9.125 dollars _____ **(c)** 15 ÷ 4 dollars _____

(d) Three friends share 25 dollars equally. How much does each friend get? How many cents are left over?

Each friend: _____ dollars Left over: _____ cents

E4 Word problems

(a) Mr Bowen buys 3 notebooks at 2.45 dollars each and one pen for 1.80 dollars. He pays with a 10-dollar note. How much change does he get?

(b) Mr Bowen buys 0.5 kg of grapes. He eats a quarter of them on the walk home, which he calls "quality control". How many kilograms of grapes arrive at his house?

E5 Now you write the question

The rules

A **real** situation — a shop, a recipe, a race, a bag on a scale. Full sentences, ending in a **question**. Use one of the **words** from E2. Give **units**. Your question must come out at the answer given.

Write a word problem for $12.60 \div 4 = 3.15$

Section F · Science 3.1 in decimals — Mr Bowen in Space

Weight and mass, from Science 3.1

weight (N) = mass (kg) × strength of gravity (N/kg)
On Earth, the strength of gravity is **10 N/kg**. Say it: **"ten newtons per kilogram"**.
Mass does not change from planet to planet. Weight does.

F1 How strong is gravity?

These are the strengths of gravity from the lesson. A bigger number pulls harder.

Place	Moon	Mars	Venus	Uranus	Earth	Neptune	Jupiter
N/kg	**1.6**	**3.7**	**8.9**	**8.7**	**10**	**11**	**25**

(a) Write the seven strengths in order of size, starting with the smallest.

_____

(b) Which pulls harder, **Venus** or **Uranus**? By how much? _____

(c) How much stronger is gravity on the **Earth** than on the **Moon**? _____

F2 Mr Bowen's space backpack

Mr Bowen's space backpack has a mass of **15 kg**. **(a)** Find its weight in each place. Estimate first. The first one is done for you.

Place	Gravity (N/kg)	Calculation	Weight (N)
Earth	10	15×10	150
Moon	1.6		
Mars	3.7		
Venus	8.9		
Jupiter	25		

(b) How much **less** does the backpack weigh on the Moon than on the Earth? _____

F3 Do heavy things fall faster?

Mr Bowen drops a **heavy** ball and a **light** ball, the same size, from 1 metre. He times each fall with a stopwatch, four times.

	Drop 1	**Drop 2**	**Drop 3**	**Drop 4**
Heavy ball (s)	0.48	0.41	0.50	0.41
Light ball (s)	0.44	0.52	0.39	0.45

Word help

conjecture	a guess that you can test
mean	add up all the values, then divide by how many there are
reaction time	how long you take to move. It is about **0.2 s**

(a) Find the **fastest** time in the table. Which ball was it? _____

(b) Make a **conjecture** from (a): which ball falls faster? _____

(c) Now test it. Find the **mean** time for each ball.

Heavy ball **Light ball**

(d) Do heavy things fall faster? Use your two means. Answer in a full sentence.

(e) The four times for the **same** ball are all different. Why? Use the words **reaction time**.

F4 Mr Bowen moves house

Mr Bowen's mass is **80 kg**. He would like to weigh exactly **296 N** without eating any less. Which place in **F1** should he move to? Show your calculation.

He has started packing. What will his **mass** be when he gets there? _____

Answer Key — Teacher Copy

A1 Put each number on the line

A on the 2nd small line · B on the 9th · C halfway between the 4th and 5th · D halfway between 5 and the 1st.
(b) 5.05, 5.2, 5.45, 5.9

D is the one to watch. A student who puts 5.05 *after* 5.2 has read it as 5.5 — the zero after the point was not said — and that is the whole of 4.1 in one mark. The line is there so they can see 5.05 is the first thing after 5. If (b) is right but the line is wrong, they ordered by the rule and did not trust the picture; that is fine.

A2 Which is bigger?

(a) > (b) > (c) < (d) > (e) = (f) <
(g)(i) six point one eight · six point two.
(ii) He is wrong. 6.2 = 6.20, and 6.20 > 6.18: two tenths is more than one tenth.

(g) is the deck's slide 3 with new numbers, and (i) is the point of it. Anyone who writes "six point eighteen" has found the cause of the mistake by themselves: English reads the digits after the point one at a time, and "eighteen" is the French *virgule dix-huit* (and the Vietnamese) leaking through. Read (i) before marking (ii).
(d) is the one that goes wrong the other way: 9.99 has more digits and more nines, and 12.3 is bigger because the whole-number part is checked first.
(f) needs *two* zeros of filling: 4.125 against 4.130.

A3 Who won?

(a) 4.08, 4.5, 4.75, 4.79, 4.8 (b) **A**
(c) 30.9, 30.95, 31.09, 31.12, 31.4 (d) **Lane 4** (e) **Lane 5** (f) **Lane 1**

Every answer here is the number that *looks* wrong. The winning jump, 4.8, is the shortest-looking number in the list; the slowest swim, 31.4, is the shortest-looking time. A student who orders by length gets both upside down.
(e) asks for *fourth*, not the deck's *third*, on purpose: with five lanes the third is the middle one from either end, so it cannot catch anyone counting from the wrong end. Fourth can. Expect Lane 2, which is fourth counted from the *slowest*. (d) is the deck's "fastest means smallest" — ask anyone who wrote Lane 1 whether a fast swimmer takes more time or less.

B1 Use a written method

(a) 8.06 (b) 18.45 (c) 8.35 (d) 5.65 (e) 7.64 (f) 0.35

(a) is the alignment check: anyone who lines up the *right-hand digits* instead of the points adds 536 + 27 and writes 5.63.
(c) has a whole number with no point at all. 6 is 6.00; students who cannot see where its point is put the 6 under the 5 and get 2.41.
(d), (e) and (f) need the zeros *written* before subtracting: 9.40, 12.00, 1.00. (f) is the deck's 1 − 0.72 with new numbers; the usual wrong answer is 0.45 (borrowing from the tenths and forgetting the hundredths).

B2 Use a mental method

3.9 + 2.5 = 4 + 2.5 − 0.1 = 6.4
7.3 − 1.9 = 7.3 − 2 + 0.1 = 5.4
6.99 + 3.45 = 7 + 3.45 − 0.01 = 10.44
12.6 − 4.98 = 12.6 − 5 + 0.02 = 7.62

The sign of the give-back is the question. When you take away *too much* you have to give some back, so the subtractions end in +. Expect 5.2 for the second row and 7.58 for the fourth — the right move with the wrong sign.
Row 4 gives back 0.02, not 0.2: 5 − 4.98 is two hundredths.
Accept any correct easier way (e.g. 3.9 + 2.5 = 3.9 + 2.1 + 0.4). The column must show a calculation, not just the answer.

C1 Point out, point back

6.4 × 5	6 × 5 = 30	64 × 5 = 320	**32**
0.7 × 8	1 × 8 = 8	7 × 8 = 56	**5.6**
2.35 × 4	2 × 4 = 8	235 × 4 = 940	**9.4**
8.15 × 2	8 × 2 = 16	815 × 2 = 1630	**16.3**

Rows 1, 3 and 4 end in a zero after the point is put back (32.0, 9.40, 16.30). Accept those, and accept the zero dropped: nothing was rounded, as the deck's own practice answer says (5.00 = 5).
Row 2 is the deck's vote (0.3 × 4) in disguise. Expect 0.56 from anyone who counts *all* the digits instead of the decimal places, and the estimate catches it: the answer must be near 8.
Estimates may differ (2.35 × 4 could be 2.5 × 4 = 10); any sensible rounding is right.

C2 The point goes straight up

(a) 2.52 (b) **1.04** (c) 0.12 (d) **1.35** (e) **3.24** (f) **1.01**
(g) 1.04 × 6 = 6.24

(b) and (f) need a 0 in the answer where a digit does not divide: 6 into 2 will not go. Expect 1.4 and 1.1, which is the deck's "2.6 forgot the 0" exactly. The inverse check in (g) catches it: 1.4 × 6 = 8.4, not 6.24.
(d) and (e) run out of digits: write 5.40 and 16.20 and keep going. Expect "1.3 remainder 2" — the remainder belongs in the hundredths.
(c) starts with 0 in the ones: 7 into 0 will not go, so the answer begins 0.

C3 One fact, five answers

(a) 102.6 (b) 10.26 (c) 102.6 (d) 3.8 (e) 0.38
(f) (a) and (c). Both have one decimal place in total, so both are 1026 with one place put back. (Or: 3.8 × 27 and 38 × 2.7 are both about 4 × 27 ≈ 100.)

Estimates: (a) 4 × 30 = 120, so 102.6 and not 10.26 or 1026. (d) is (a) undone, and (e) is (b) undone — a student who sees that has the inverse calculation without being told.
Anyone who multiplies 38 × 27 again has not read the instruction; it is the English of "do not" and worth one line.

D1 Decimals as fractions

(a) $\frac{7}{10}$ (b) $\frac{3}{100}$ (c) $\frac{31}{100}$ (d) $\frac{6}{10}$ (or $\frac{3}{5}$) (e) 0.08 (f) 0.51

(b) and (e) are the same trap in two directions: 0.03 = $\frac{3}{10}$ and $\frac{8}{100}$ = 0.8. Both come from reading the digit and ignoring the column. Send them back to the word-help line: the *last* digit's column is the bottom number. Simplifying is not required.

D2 Half of, a quarter of, a tenth of

a quarter of 36, 9 · a tenth of 70, 7 · 2.8 × 0.25, 0.7 · 4.5 × 0.1, 0.45

Rows 4 and 5 are the English-to-calculation direction, and the column is "calculation": accept 2.8 ÷ 4 as well as 2.8 × 0.25, but point out that the deck wrote it with × because "of" means ×.
Row 5 is the one to watch: a tenth of 4.5 is 0.45, not 0.5 or 4.4.
Row 4 needs the 0 in the ones: 4 into 2 will not go, so 0.7 (C2(c) again).

D3 Multiplying by a decimal

(a) 4 × 3 = 12 (b) 15 × 2 = 30 (c) 6 × 4 = 24 (d) 9 × 7 = 63 (e) 12 × 5 = 60 (f) 8 × 6 = 48

(c) and (e) divide by **100**, not 10. Expect 60 × 4 = 240 and 120 × 5 = 600: the student divided by 10 for a hundredth.
Every answer is smaller than the number it started from. Until this unit every multiplication the class has done made the number bigger (the deck's slide 6); it is worth asking aloud whether anyone noticed. D5(d) tests it.

D4 Times 9, times 11, and partitioning

(a) 46 − 4.6 = 41.4 (b) 23 + 2.3 = 25.3
(c) 6 × 7 + 0.2 × 7 = 42 + 1.4 = 43.4 (d) 3 × 5 + 0.8 × 5 = 15 + 4 = 19

(a) needs a decimal subtraction with a hidden zero: 46.0 − 4.6. Expect 42.4 from 46 − 4 + 0.6.
(d) is partitioning landing on a whole number: 0.8 × 5 = 4.0. Students who write 0.4 have counted the decimal places in the question as if they were in the answer — 8 × 5 = 40, one place back, 4.0.
Accept the column method for (c) and (d), but ask for the partition: the deck taught it as the faster way.

D5 Mr Bowen's homework

(a) **wrong** — 2.8. One decimal place in, one decimal place out; the estimate is 0.5 × 7 ≈ 3.
(b) **wrong** — 5.09 < 5.1. Fill with zeros: 5.09 against 5.10.
(c) **right.**
(d) **wrong** — 15. Multiplying by a decimal less than 1 makes it smaller: 5 × 3.

One trap each from C1, A2, C2 and D3, in that order. Whichever one a student gets wrong, send them to that question rather than explaining again.
(c) is right on purpose, and it is the one with the most room for doubt: 7.2 ÷ 6 gives 1.2 and a student who expects every division in this unit to need a 0 will "correct" it to 1.02.

E1 Two-digit numbers

(a) 87 (58 × 15 = 870; estimate 6 × 15 = 90)
(b) 3.2 (13 table: 13 × 3 = 39, remainder 2.6 = 13 × 0.2)

(a) lands on a whole number, 87.0; expect 8.70 from a student who put back two places because the question had "two numbers". The estimate, 90, is the fastest correction.
For (b) the table is the method — look for it written down. A student with no table and the right answer probably knew it; a student with no table and a wrong answer needs to be told to write the table, not shown the division.

E2 From English to a calculation

(a) 5.00 − 3.65 = 1.35 dollars (b) 1.20 × 6 = 7.20 dollars
(c) 13.50 ÷ 3 = 4.50 dollars (d) 2.4 − 0.85 = 1.55 kg
(e) 9.1 − 2.6 = 6.5

Mark the calculation, not the answer. This is the English question, and the numbers were chosen to be easy.
(c) is the deck's slide 20 trap: *Mr Bowen and two friends* is **three** people. 13.50 ÷ 2 = 6.75 is an English

error, not an arithmetic one — make the student count the people out loud.
(e) puts the numbers in the opposite order to the sentence. A student who writes 2.6 − 9.1 has read left to right; ask what "subtract 2 from 9" is in the word-help box.
(a) and (b): accept 5 − 3.65 and 1.2 × 6, but the answers are money and should be 1.35 and 7.20 (E3).

E3 Money has two decimal places

(a) 4.80 (b) 9.13 (c) 3.75
(d) 25 ÷ 3 = 8.333 . . ., so **8.33** dollars each, and 25.00 − 24.99 = 0.01, so **1 cent** is left over.

(b) is rounding: the 5 in the thousandths rounds the 2 up. Expect 9.12 (cut off, not rounded) and 9.125 (left alone).
(d) is the deck's slide 10 with 25 instead of 10. "How many cents" is the English: some will write 0.01, which is right in dollars and should be turned into the word the question used.

E4 Word problems

(a) 3 × 2.45 = 7.35; 7.35 + 1.80 = 9.15; 10.00 − 9.15 = **0.85** dollars.
(b) A quarter of 0.5 is 0.5 ÷ 4 = 0.125 kg. 0.5 − 0.125 = **0.375** kg.

(a) is three steps, and the usual slip is to stop after two and give 9.15 as "the change". Ask what the question asked for.
(b) is the deadpan one and the hardest item in E. 0.500 ÷ 4 needs two zeros written to finish, and the question asks what *arrives*, not what he ate. Expect 0.125 as the final answer; that is the grapes Mr Bowen ate. Accept 0.5 × 0.75 if anyone finds it, but nobody is expected to.

E5 Now you write the question

Check, in order: a real situation; full sentences; ends in a question mark; uses a word from E2; gives units; and the student's own question really comes out at the answer.

It wants *share equally* with **four** people. A good answer: "Mr Bowen and three friends share a 12.60-dollar pizza equally. How much does each person pay?"
Watch for "Mr Bowen and four friends", which is five, and is the E2(c) error made by the student who wrote the question. That one is worth reading aloud to the class.
The other common failure is a question whose answer is 50.40: "each of 4 friends has 12.60 dollars — how much altogether?" Read it back to them.

F1 How strong is gravity?

(a) 1.6, 3.7, 8.7, 8.9, 10, 11, 25
(b) **Venus**, by 8.9 − 8.7 = 0.2 N/kg. (c) 10 − 1.6 = 8.4 N/kg.

(a): 8.7 and 8.9 are the pair to check, and 10 and 11 after the two eights. A student ordering by digit count puts 10, 11 and 25 *before* 8.9.
(c) needs 10.0 − 1.6; expect 9.6 or 8.6 from anyone who did not write the zero.

F2 Mr Bowen's space backpack

Moon 15 × 1.6 = 24 N · Mars 15 × 3.7 = 55.5 N · Venus 15 × 8.9 = 133.5 N · Jupiter 15 × 25 = 375 N
(b) 150 − 24 = **126** N

Estimates: Moon 15 × 2 = 30; Mars 15 × 4 = 60; Venus 15 × 9 = 135. A student who writes 2.4 or 240 for the Moon has the digits right and the point wrong, and the estimate of 30 settles it.
Venus invites ×9: 15 × 8.9 = 15 × 9 − 1.5 = 133.5 is D4 used without being asked — worth praising aloud.
The mass on every row is still 15 kg; F4's last line asks that, and it is worth saying here if anyone wrote a different mass for the Moon.

F3 Do heavy things fall faster?

(a) 0.39 s, the **light** ball. (b) "The light ball falls faster."
(c) Heavy: 0.48+0.41+0.50+0.41 = 1.80, 1.80÷4 = **0.45** s. Light: 0.44+0.52+0.39+0.45 = 1.80, 1.80 ÷ 4 = **0.45** s.
(d) "No. Both balls have a mean time of 0.45 s, so they fall at the same speed."
(e) "Mr Bowen's reaction time is about 0.2 s, so he starts and stops the stopwatch a little early or late each time."

This is the question the packet is built around. The fastest single time belongs to the light ball, so the conjecture in (b) is *supposed* to be wrong. That is what a conjecture is for: the mean tests it and throws it out. Expect some students to "fix" (b) after doing (c); ask them to leave it and say in (d) that it was wrong.
(d) answers the question Science 3.1 opened on (Galileo, the hammer and the feather), and it lands on the same answer the deck gave — in decimals, from numbers. It is worth putting both means on the board next to each other.
(e) is slide 43: the times differ by up to 0.11 s, about half a reaction time. Accept any sentence that blames the person with the stopwatch and uses "reaction time".

F4 Mr Bowen moves house

296 ÷ 80 = 29.6 ÷ 8 = 3.7 N/kg, which is **Mars**. (Or try each place: 80 × 3.7 = 296.)
His mass will be **80 kg** — mass does not change.

Either method is fine; trial with F1's list is perfectly good mathematics and probably the more common route. 296 ÷ 80 is the deck's "Mr Bowen in Space" run backwards, and some students will remember 296 N from the lesson — that also counts.
The second line is the science, and it is the one to discuss rather than mark. Mass does not change, so it needs no calculation at all, and some students will distrust that: expect 296, or 80 × 3.7 worked out again. Mercury (also 3.7) is left out of F1 so that the answer is a single place.